Limit-Hitting Exciting Effects: Modeling Jump Dependencies in Price-Limit Markets

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March 30, 2026

Based on Chen & Qi (2024)

Introduciton

Motivation: Price-Limit Markets and Jump Behavior

- Many stock markets worldwide implement **daily price-limit rules** (e.g., China, Japan).
- Traditional models implicitly assume:
  - Jumps are independent across time
  - No distinction between limit-hitting (LH) and small jumps
- Empirical evidence shows:
  - LH events attract strong investor attention
  - LH events significantly increase the probability of future jumps
  - Return and volatility clustering around LH days
- **Motivating Question:**
  Do limit-hitting events generate **systematic, mutually-exciting jump interactions**?

Motivation: Limitations of Standard Jump Models

- Standard SVJ frameworks do not capture:
  - Behavioral differences between LH and small jumps
  - Interaction between jumps (self- or cross-excitation)
  - Microstructure distortions under price limits
- LH events are fundamentally different:
  - Larger magnitude and higher informational content
  - Cause concentrated trading pressure and volatility bursts
  - Have predictive power for future jumps
- **Research Gap:**
  A model is needed to explicitly separate LH vs. small jumps and allow them to interact dynamically.

Motivation: Microstructure Basis for LHE Effects

Kim & Rhee (1997) identify three major hypotheses induced by daily price limits:

1. **Delayed Price Discovery**
   - Price limits slow information incorporation, leading to delayed adjustments and increasing the likelihood of future jumps.
2. **Volatility Spillover**
   - LH events create severe order imbalance, leading unexecuted orders to spill into later sessions and amplify the volatility.
3. **Trading Interference**
   - Constrained price movement forces impatient traders to trade at unfavorable prices, generating abnormal volume and noise.

Motivation: Why a New Model Is Needed

- LHE effects create clustered and interdependent jumps, directly violating the serial-independence assumption in standard SVJ models.
- Ignoring LHE produces biased estimates of jump risk and leads to poor model fit in price-limit markets.
- LH events have distinct jump characteristics and dynamically interact with small jumps— features that classical models cannot capture.
- **This paper** develops the first model that
  - explicitly incorporates price-limit rules,
  - separates LH from small jumps, and
  - models mutually-exciting jump dynamics via a multivariate Hawkes process.

Research Questions

Research Questions

- How do LH jumps differ from small jumps in size distribution and informational content?
- Do LH events generate **excitatory effects** that increase future jump arrivals?
- Can we model jump interactions using a mutually-exciting (Hawkes) framework?
- How do LH-induced jump dynamics vary across industries or firm characteristics?

Model Setup

Model Overview

The model consists of three main components:

- A **stochastic volatility (Heston)** process for continuous returns.
- **Four jump types** representing limit-hitting and small jumps: pl^+, pl^-, y^+, y^-.
- A **mutually-exciting Hawkes structure** that governs how jumps trigger subsequent jumps (self-excitation and cross-excitation).

This structure captures the clustered, asymmetric, and contagious jump behavior caused by daily price-limit mechanisms.

Price and Variance Dynamics

Price process:

$$dp_t = \mu\, dt + \sqrt{v_t}\, dW_t^p + \xi_t\, dN_t,$$

Variance process:

$$dv_t = \kappa(\theta - v_t)\, dt + \sigma_v \sqrt{v_t}\, dW_t^v.$$

Interpretation:

- μ: drift of the return process.
- v_t: instantaneous variance level (CIR dynamics).
- κ, θ: speed of mean reversion and long-run mean of v_t.
- σ_v: volatility of volatility.
- $\mathrm{corr}(dW_t^p, dW_t^v) = \rho$.
- $\xi_t dN_t$: jump component of returns. Different jump types have different sizes ξ_t and arrival processes N_t.

Jump Components and Jump Size Distributions

Jump decomposition:

$$\xi_t dN_t = \sum_i \xi_t^i \, dN_t^i, \qquad i \in \{pl^+, pl^-, y^+, y^-\}$$

Jump arrivals:

$$\mathbb{P}(dN_t^i = 1) = \lambda_t^i,$$

where λ_t^i is the intensity of jump in dimension i.

Interpretation:

- pl^+, pl^-: upper and lower price-limit-hitting jumps.
- y^+, y^-: positive and negative small jumps.

Jump size distributions:

$$\begin{cases} \xi_t^{pl^+} \sim \mathrm{EXP}(\mu_{pl^+}), \qquad \xi_t^{pl^-} \sim -\mathrm{EXP}(\mu_{pl^-}), \\ \xi_t^{y^+}, \xi_t^{y^-} \sim \mathcal{N}(\mu_p, \sigma_p). \end{cases}$$

Mutually-Exciting Jump Intensities

Jump intensity for type i (MHP):

$$\lambda_t^i = \lambda_0^i + \sum_j \int_0^t \alpha_{j,i} e^{-\beta_{j,i}(t-s)} N_j ds, \qquad i, j = \{pl^+, pl^-, y^+, y^-\}.$$

Components:

- λ_0^i: baseline jump intensity for type i.
- $\alpha_{j,i}$: incremental impact on the intensity of type-i after a jump of type-j. (It captures self and cross excitation.)
- $\beta_{j,i}$: decay rate of the incremental intensity.

Interpretation:

- Past jumps raise future intensities $\rightarrow$ **jump clustering**.
- Cross-dimension interactions represent LHE effects under price limit rules.

Excitation Effects in the Hawkes Jump Model

Self-excitation ($i = j$):

- A jump increases the probability of another jump of the same type.
- Creates clustering such as $pl^- \to pl^-$ during sell-offs.

Cross-excitation ($i \neq j$):

- A jump of type j raises the intensity of a different jump type i.
- Explains spillover effects such as $pl^- \to y^-$ or $pl^+ \to y^+$.

Why important?

- Captures LHE effects: limit-hitting events trigger additional jumps.
- Generates volatility amplification and directional contagion.

Discrete-Time Representation for Estimation

To estimate the model using daily data, the continuous-time SVJ system is discretized using Euler's method with time step $\Delta = \frac{1}{252}$:

Discrete price equation:

$$p_t = p_{t-1} + \mu\Delta + \sqrt{v_t \Delta}\, \varepsilon_t^p + \xi_t J_t.$$

Discrete variance equation:

$$v_t = v_{t-1} + \kappa(\theta - v_{t-1})\Delta + \sqrt{v_{t-1}\Delta}\, \sigma_v \varepsilon_t^v.$$

Shock structure:

- $\varepsilon_t^p, \varepsilon_t^v \sim \mathcal{N}(0,1)$ with $\mathrm{corr}(\varepsilon_t^p, \varepsilon_t^v) = \rho$.
- $J_t \sim \mathrm{Bernoulli}(\lambda_t^i)$ indicates whether a jump occurs at time t.
- Jump sizes ξ_t follow the same distribution as in the continuous-time model.

Model Estimation Setup

We estimate the model using a **Bayesian MCMC** procedure.

Static parameter vector:

$$\Theta = \{\mu, \kappa, \theta, \sigma_v, \rho, \mu_p, \sigma_p, \mu_{pl^+}, \mu_{pl^-}, \lambda_0, \alpha, \beta\}$$

- α, β: 4×4 matrices for Hawkes excitation effects.
- λ_0: 4×1 baseline intensities for $\{pl^+, pl^-, y^+, y^-\}$.

Latent variables:

$$\Upsilon_t = \{v_t,\ \Xi_t,\ J_t\}$$

- v_t: variance path.
- Ξ_t: jump sizes (LH sizes latent, small jumps latent).
- J_t: jump indicators (LH jumps observed, others require inference).

Bayesian Posterior and MCMC Procedure

Joint posterior distribution:

$$P(\Theta, \Upsilon_t \mid p_t) \propto P(p_t \mid \Theta, \Upsilon_t)\, P(\Upsilon_t \mid \Theta)\, P(\Theta)$$

Why MCMC is required:

- The likelihood has no closed-form due to:
  - latent jumps and latent volatility,
  - path-dependent Hawkes intensities,
  - non-Gaussian jump size distributions.
- MCMC allows sequential sampling of:
  - static parameters Θ,
  - latent volatilities v_t,
  - jump sizes and indicators (Ξ_t, J_t).
- Provides full uncertainty quantification for excitation effects.

Empirical Analysis

Data Overview and PLH Event Statistics

Data Description

- Sample: 3,646 A-share stocks from the CSMAR database.
- Period: Jan 2007 – Dec 2021 (daily closing prices).
- PLH events identified following exchange rules.

Data Filtering

- Exclude firms without complete equity-division reform.
- Exclude stocks listed after 2020 (insufficient trading history).
- Remove stocks with no PLH events or abnormal price records.

PLH events are not rare: the average PLH frequency per year is 3.14%.

Table 1
Descriptive statistics of price limit hittings.

Variable	# of Stocks	Mean	Std. dev.	Min	Max
Total PLH	3646	94.14	64.58	4	301
PLH per year	3646	7.91(3.14%)	5.73(2.27%)	0.55(0.22%)	26.62(10.56%)

Note: This table reports the descriptive statistics of PLH events. Total PLH stands for total PLH events during the data period, while PLH per year is the PLH events observed per trading year. The percentage in the bracket of the last row stands for the frequency of PLH events.

Model Specifications and Benchmark

Table 2
Model specifications.

Model	Jump Specifications
$\mathcal{M}_1$ (Heston, 1993)	No jump components
$\mathcal{M}_2$ (Bates, 1996)	No distinction between LH jumps and small jumps; jump sizes are modeled by a Normal distribution; jump intensity is constant
$\mathcal{M}_3$ (Kou and Wang, 2004)	No distinction between LH jumps and small jumps; jump sizes are modeled by double exponential distributions; jump intensity is constant
$\mathcal{M}_4$ (NEW)	LH jump sizes are modeled by double exponential distributions; small jump sizes are modeled by a Normal distribution; jump intensities are constant
$\mathcal{M}_5$ (NEW)	LH jump sizes are modeled by double exponential distributions; small jump sizes are modeled by a Normal distribution; jump intensities are modeled by a mutually exciting process

Benchmark Model: NGARCH (Engle & Ng, 1993)

- Classic nonlinear volatility model (clustering + leverage).
- Cannot capture discontinuous jumps or excitation..

Illustration of Jump and Volatility Components

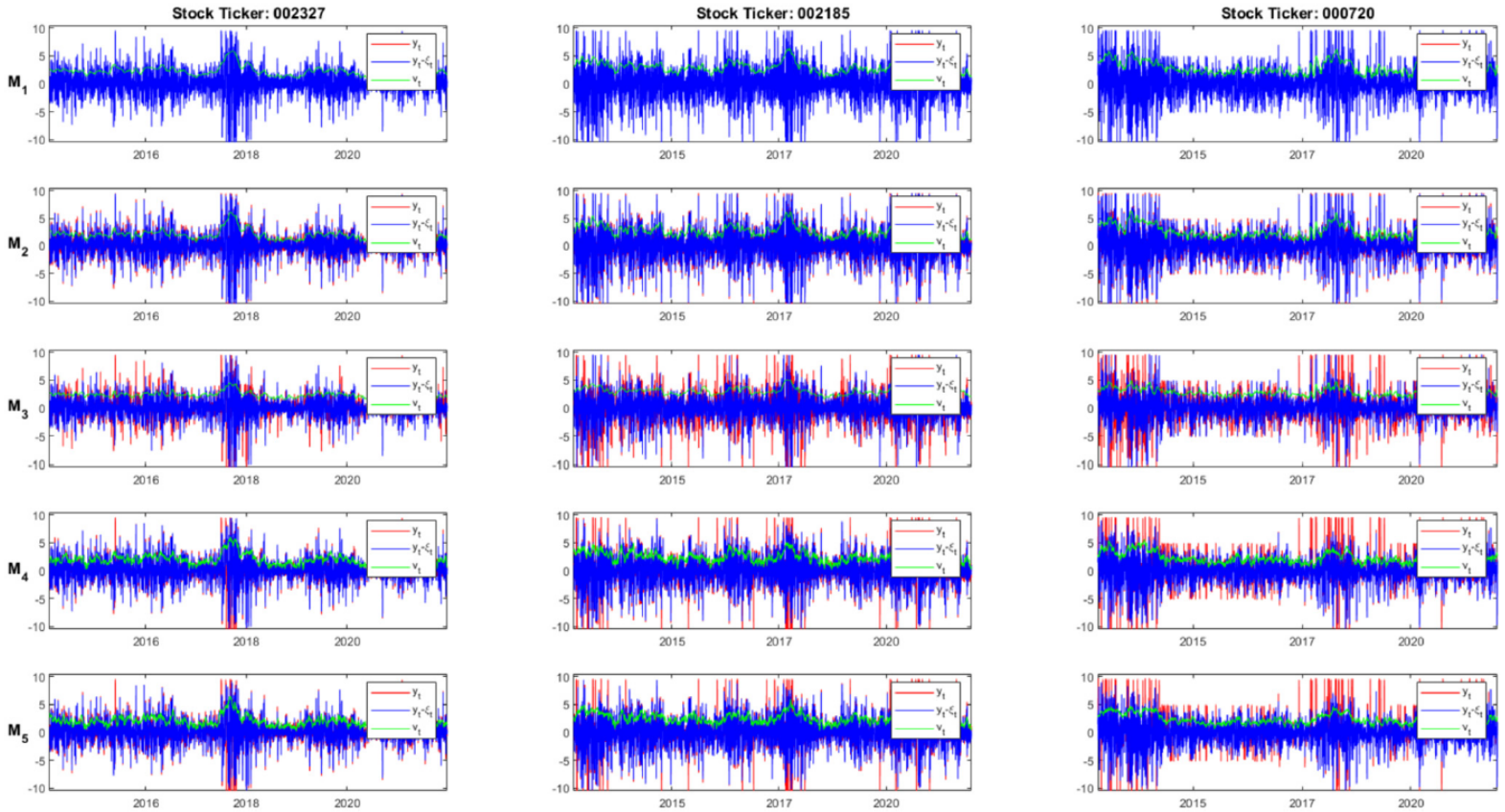


Fig. 1. Stock daily returns, returns minus estimated jumps and estimated volatility for the three A-share stocks. Jumps and volatility are estimated by different models listed in Table 2. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

- **Stock selection:** Sort all A-share stocks by the number of LH events and choose stocks at the 10th, 50th, and 90th percentiles.
- **Key insight:** M_1–M_2 rely on volatility to explain returns, and frequent LH events require separating jump components.

Model Fit Results: DIC and LLK

Table 3
DIC and LLK of different models.

Model	002327	002185	000720	All	002327	002185	000720	All
PLH(%)	0.59%	2.35%	6.20%	3.07%	0.59%	2.35%	6.20%	3.07%
	DIC				LLK			
NGARCH	-	-	-	-	0.80	0.97	0.81	0.85(0%)
$\mathcal{M}_1$	1.00	1.00	1.00	1(0%)	1.00	1.00	1.00	1(0%)
$\mathcal{M}_2$	0.94	0.98	0.96	0.99(%)	1.03	1.02	1.01	1.02(0.5%)
$\mathcal{M}_3$	0.96	0.99	0.93	0.99(9.2%)	1.00	1.01	1.02	1.01(11.5%)
$\mathcal{M}_4$	**0.89**	0.91	0.92	0.98(20.6%)	1.04	1.00	1.02	1.03(16.3%)
$\mathcal{M}_5$	0.91	**0.89**	**0.93**	0.97(69.3%)	**1.06**	**1.02**	**1.05**	1.08(71.6%)

Note: This table presents the Deviance Information Criterion (DIC) and log-likelihood (LLK) of three example stocks with ID: 002327, 002185, and 000720. The second row, PLH(%), presents the percentage of price-limit-hitting events of three stocks during the data period. The percentage is 3.07% for all stocks on average. Column "All" represents the average value of all stocks (and the percentage of stocks that have the lowest DIC and the highest LLK using the corresponding model). The values of DIC and LLK of $\mathcal{M}_1$ are set as 1, while those for other models are represented as ratios relative to $\mathcal{M}_1$.

- M_5 outperforms all competing models, delivering the best fit (lowest DIC / highest LLK) for **around 70% of stocks**.
- Its performance improves further in groups with more PLH events, showing that incorporating LHE dynamics is essential in price-limit markets.

Distribution of Static Parameters

Posterior distributions of return and variance process parameters

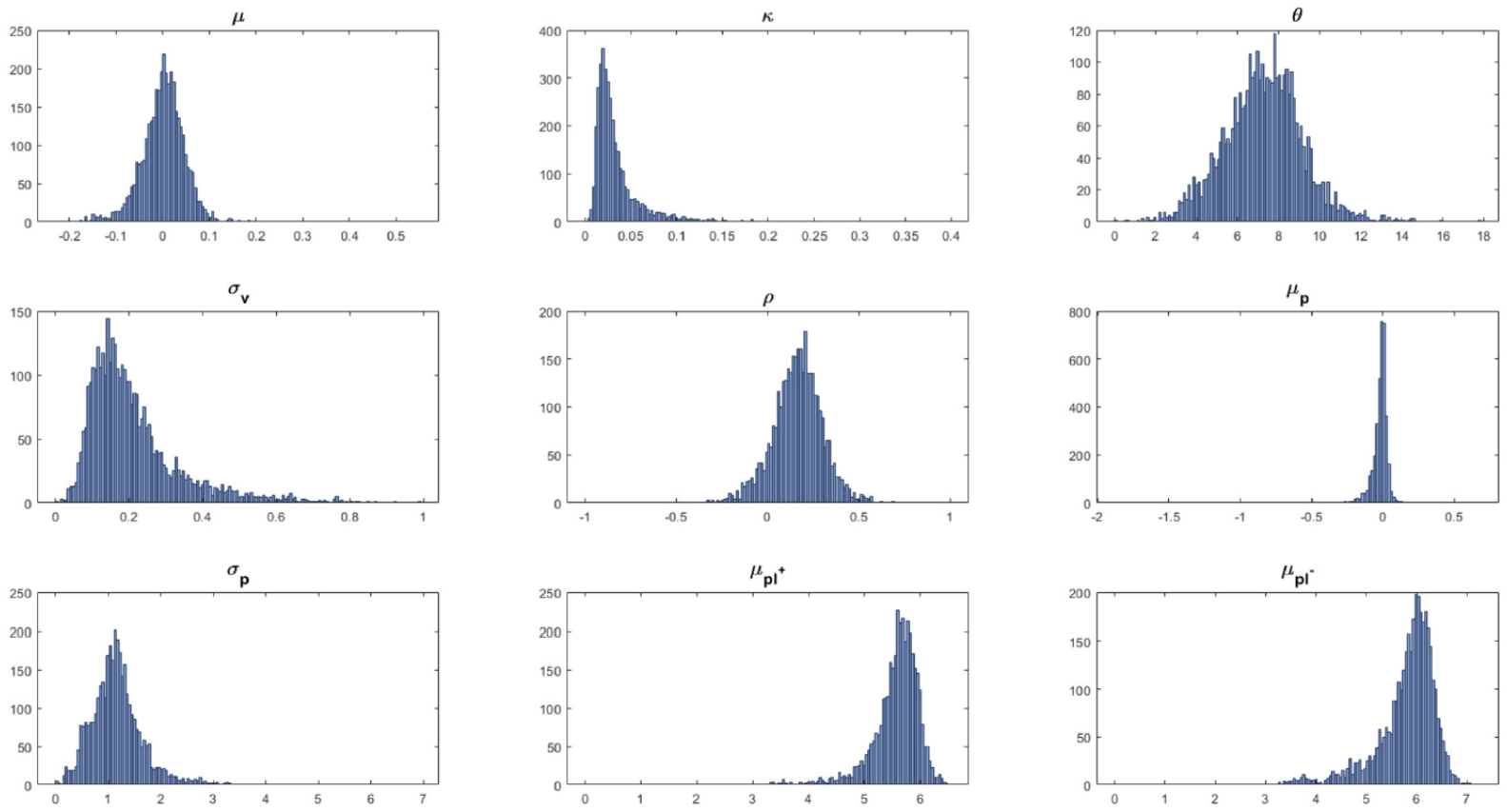
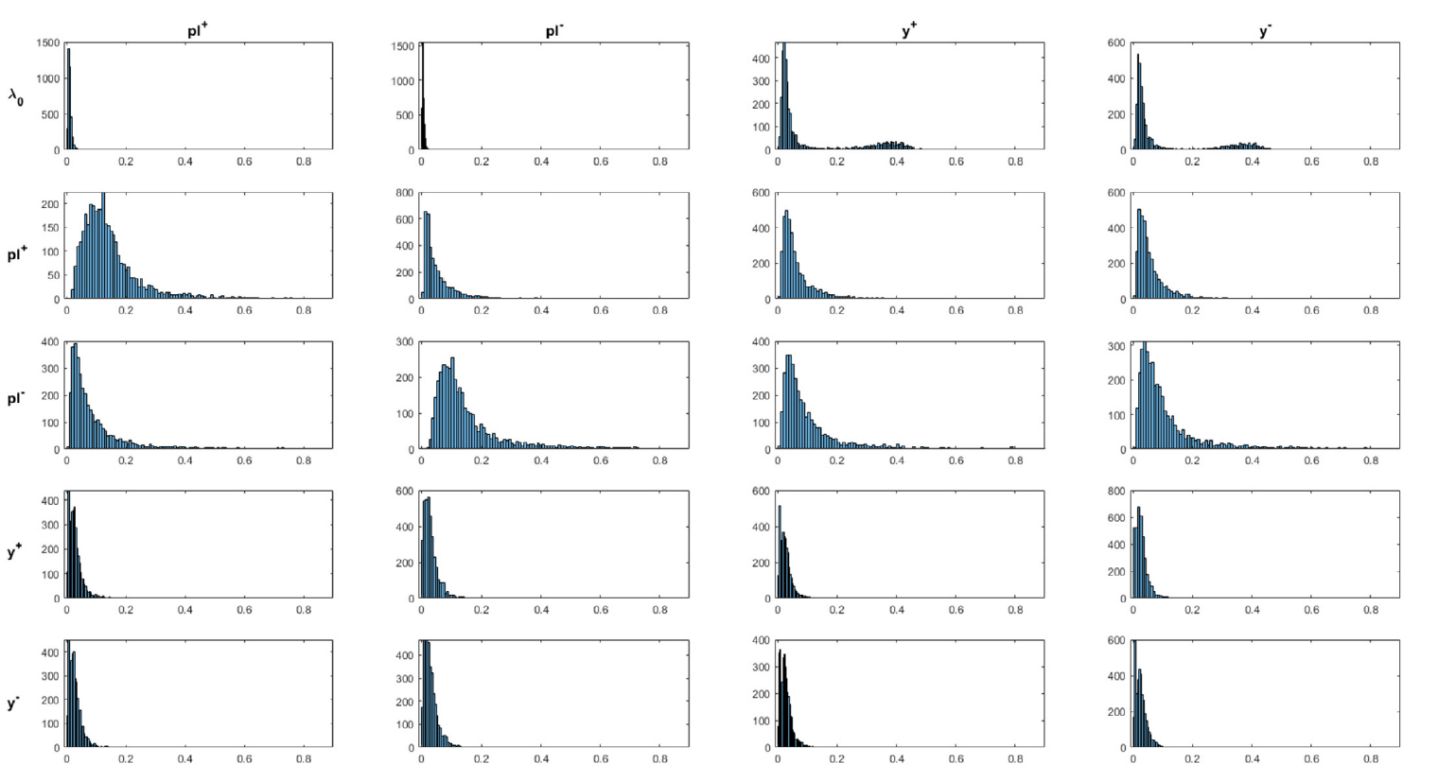


Fig. 2. Histograms of static parameters in the return and variance process. The x-axis represents the posterior means of the estimated parameters while the y-axis represents the number of companies.

Distribution of Hawkes Parameters

Market-wide asymmetry and heterogeneity in jump–excitation effects



Histograms of Hawkes excitation parameters (λ_0, α) show strong asymmetry between LH and small jumps, indicating heterogeneous clustering behaviors across firms.

Significance of Hawkes Parameters

Significance of estimated parameters.

	$j \backslash i$	pl^+	pl^-	y^+	y^-
λ_0^i		100.0%	100.0%	99.8%	99.8%
$\alpha_{j,i}$	pl^+	95.3%	67.3%	21.1%	19.4%
	pl^-	75.5%	92.8%	32.1%	35.0%
	y^+	4.0%	2.9%	2.6%	3.5%
	y^-	3.0%	4.5%	3.2%	2.8%
$\beta_{j,i}$	pl^+	96.9%	85.2%	99.5%	98.8%
	pl^-	86.7%	96.6%	99.0%	99.3%
	y^+	99.6%	99.8%	99.2%	98.5%
	y^-	99.8%	99.8%	98.3%	99.3%

- Although LH events occur less frequently, they have a **significant impact on future jump intensities**.
- ULH and LLH exhibit **strong self-exciting effects**: for about **65%–95%** of firms, an LH event significantly increases the probability of subsequent LH events.
- LLH events are more likely than ULH to trigger future **small jumps**, consistent with the higher significance of $\alpha_{pl^- \to y^\pm}$.

Jump Clustering and Firm Characteristics: Firm Size

Panel A. Correlations between estimated parameters and firm sizes

	j \ i	pl^+	pl^-	y^+	y^-
λ_0^i		-0.080***	-0.016	0.093***	0.092***
$\alpha_{j,i}$	pl^+	0.059***	0.084***	0.097***	0.110***
	pl^-	0.147***	0.063***	0.099***	0.093***
	y^+	-0.062***	-0.039**	-0.087***	-0.082***
	y^-	-0.064***	-0.051***	-0.094***	-0.082***
$\beta_{j,i}$	pl^+	0.104***	0.042***	-0.041***	-0.005
	pl^-	0.008	0.075	-0.027	-0.033**
	y^+	0.002	0.011	-0.017	-0.04**
	y^-	-0.002	-0.007	-0.04***	-0.017

- Larger firms exhibit **fewer LH events** due to higher liquidity and lower volatility.
- However, once LH events occur, large firms show **stronger self- and mutual-excitation**: LH events significantly increase future LH and small jumps.
- LH events in large caps are more **unexpected**, attracting greater investor attention and amplifying jump clustering.

Jump Clustering and Firm Characteristics: Turnover

Panel B. Correlations between estimated parameters and turnover ratio

λ_0^i		0.354***	0.312***	0.150***	0.155***
$\alpha_{j,i}$	pl^+	0.148***	0.062***	0.002	0.025
	pl^-	0.115***	0.188***	0.158***	0.173***
	y^+	0.034**	0.105***	0.018	0.019
	y^-	0.02	0.077***	0.019	-0.006
$\beta_{j,i}$	pl^+	0.022	0.076***	0.02	0.016
	pl^-	0.210***	0.325***	-0.009	-0.014
	y^+	-0.012	0.019	-0.003	-0.013
	y^-	-0.019	-0.005	0.014	-0.018

- Turnover (investor-sentiment proxy) is **positively correlated** with LH frequencies.
- LH jump–exciting effects are **stronger for high-turnover firms**, indicating that LH events stimulate sentiment and trigger more future jumps.
- Small jump excitation is weakly or inconsistently related to turnover, implying sentiment affects large jumps more than small ones.

Jump Clustering and Firm Characteristics: Returns

Panel C. Correlations between estimated parameters and returns

λ_0^i		0.036**	-0.057***	-0.128***	-0.150***
$\alpha_{j,i}$	pl^+	-0.212***	-0.175***	-0.181***	-0.231***
	pl^-	-0.158***	-0.172***	-0.159***	-0.151***
	y^+	0.011	-0.035**	0.024	0.033**
	y^-	0.023	0.005	0.056***	0.091***
$\beta_{j,i}$	pl^+	-0.121***	-0.055***	-0.012	0.057***
	pl^-	-0.106***	-0.115***	0.012	0.055***
	y^+	-0.005	-0.002	0.01	0.005
	y^-	0.006	0.021	0.02	0.028*

- Upper-limit (lower-limit) jump intensity is positively (negatively) related to returns, though the magnitude is small.
- Frequency of small jumps is **negatively associated** with returns.
- The strongest effect: LH incremental intensities ($\alpha_{pl^\pm,i}$) are **significantly and negatively related to returns**: higher-return firms exhibit weaker LH clustering.
- Decay speeds ($\beta_{pl^\pm,i}$) show a similar negative relation, suggesting that LH effects dissipate faster for higher-return firms.

Clustering Index and Preliminary Analysis

Conditional Jump Probabilities After LH Events

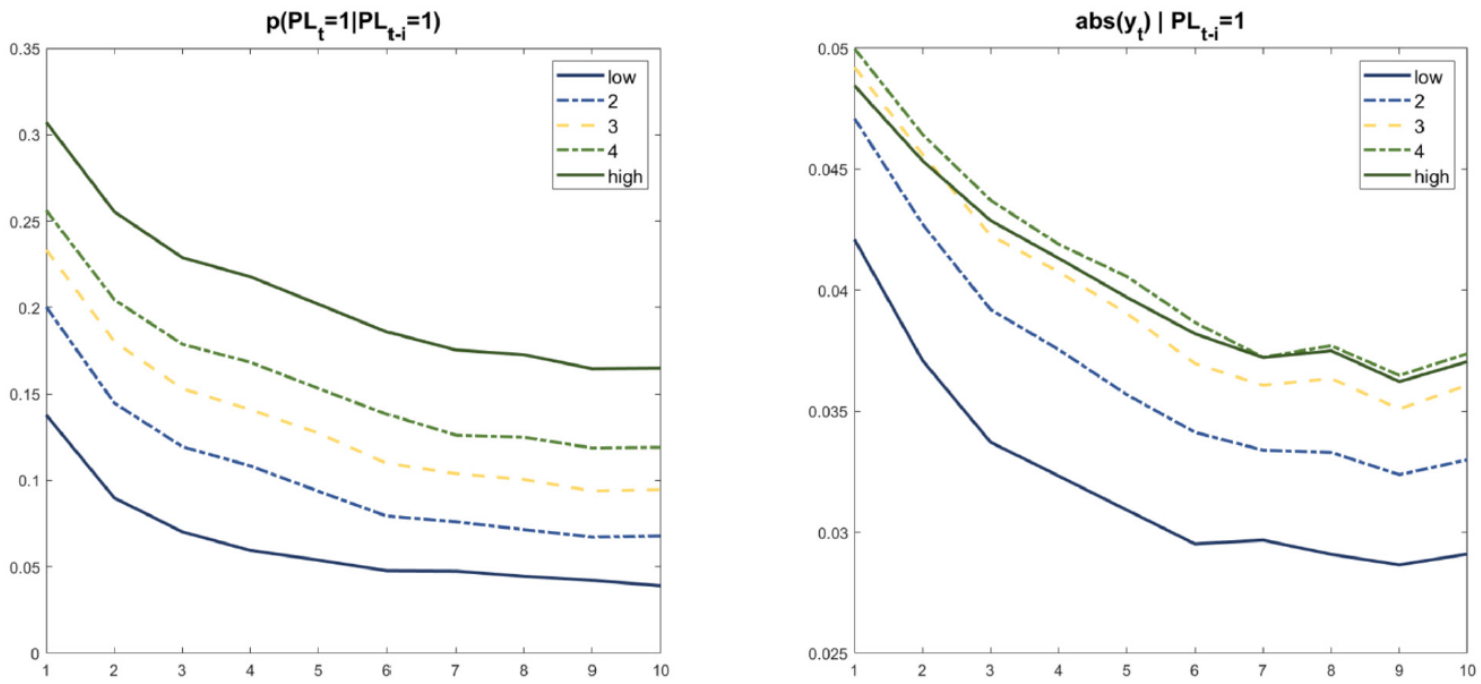


Fig. 4. Realized behaviors of limit hits. We sort all stocks into five portfolios according to the level of LHE effects. The left figure presents the empirical conditional probability of PLH events at time t given a PLH event occurred at time $t-i$, $i = 1, 2, ..., 10$. The right figure presents the mean absolute returns at time t given a PLH event occurred at time $t-i$. When calculating these returns, we exclude returns that hit price limits.

LH events sharply increase the probability of future jumps, confirming the mutually-exciting structure in real data.

Clustering Index: Definition and Insights

Clustering Index (CI):

$$CI_{j,i} = \frac{\alpha_{j,i}}{\beta_{j,i}}$$

- Measures **strength** and **persistence** of jump–excitation.
- Higher CI $\rightarrow$ larger α (impact) and smaller β (slower decay).

Empirical Findings

- Strong LH $\rightarrow$ LH and LH $\rightarrow$ small-jump clustering.
- Finance sector exhibits the strongest small-jump clustering.
- SOE firms show weaker excitation effects (more regulated trading).

Overview of Clustering Patterns

- LH events consistently generate strong **self-exciting** effects.
- LH events also cause **cross-excitation** to small negative jumps.
- Small jumps rarely excite future jumps.
- Excitation patterns vary with industry structure and ownership.

Annual LHE Effects: Variable Construction

Amplified clustering (A):

$$CI^{pl^+,A} = CI_{pl^+ \to pl^+} + CI_{pl^+ \to y^+}$$

Reversal clustering (R):

$$CI^{pl^+,R} = CI_{pl^+ \to pl^-} + CI_{pl^+ \to y^-}$$

Interpretation

- **A-effect**: LH event causes more jumps in the same direction.
- **R-effect**: LH event predicts opposite-direction jumps.
- Annual aggregation reduces noise and reveals persistent firm-level behaviors.

Cross-Sectional Patterns of Clustering Index

Table 7
Mean estimates of clustering index.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	#	λ_0^{pl}	λ_0^{y}	$CI_{pl^+,R}$	$CI_{pl^-,R}$	$CI_{pl^+,A}$	$CI_{pl^-,A}$	$CI_{y^+,R}$	$CI_{y^-,R}$	$CI_{y^+,A}$	$CI_{y^-,A}$
Panel A. Mean Estimates for All Samples											
All	3646	0.0175	0.2262	0.2155	0.1542	0.3479	0.4058	0.0979	0.0664	0.0600	0.0872
Panel B. Mean Estimates for Different Industries											
Finance	108	0.0177	0.2540	0.2225	0.1601	0.3646	0.3453	0.1525	0.1139	0.0807	0.1258
Public Utility	649	0.0187	0.2140	0.2354	0.1481	0.3600	0.4068	0.0980	0.0661	0.0585	0.0844
Properties	185	0.0150	0.1115	0.2304	0.1599	0.3716	0.4312	0.0773	0.0602	0.0487	0.1094
Conglomerates	83	0.0164	0.1407	0.2404	0.1797	0.3393	0.4293	0.0749	0.0492	0.0684	0.0678
Industrials	2443	0.0177	0.2454	0.2075	0.1520	0.3419	0.4026	0.0995	0.0666	0.0609	0.0862
Commerce	178	0.0141	0.1792	0.2218	0.1869	0.3554	0.4452	0.0747	0.0513	0.0495	0.0740
Panel C. Mean Estimates for SOE and PE											
SOE	1125	0.0150	0.1385	0.2409	0.1621	0.3363	0.4464	0.0802	0.0549	0.0506	0.0769
PE	2521	0.0187	0.2675	0.2042	0.1507	0.3531	0.3877	0.1059	0.0716	0.0642	0.0918

Note: This table presents the mean estimates of the clustering index. Panel A presents the mean values across all stocks. Panel B presents the corresponding mean estimates for different industries. The CSMAR classifies stocks into six industry groups: finance, public utility, properties, conglomerates, industrials, and commerce. Panel C shows the corresponding values for state-owned enterprises (SOE) and private enterprises (PE). The number of stocks is given in column (1); $\lambda_0^{pl} = \lambda_0^{pl^+} + \lambda_0^{pl^-}$ represents the base-line intensity for limit hits; $\lambda_0^{y} = \lambda_0^{y^+} + \lambda_0^{y^-}$ represents the base-line intensity for small jumps. We focus on two kinds of clustering effect: the amplified jump clustering effect (denoted by A) describes the situation that a jump will lead to subsequent jumps in the same direction, and the reversal jump clustering effect (denoted by R) describes the situation that a jump will lead to subsequent jumps in the opposite directions. Columns (4) to (11) show $CI_{pl^+,pl^-} + CI_{pl^+,y^-}$, $CI_{pl^-,pl^+} + CI_{pl^-,y^+}$, $CI_{pl^+,pl^+} + CI_{pl^+,y^+}$, $CI_{pl^-,pl^-} + CI_{pl^-,y^-}$, $CI_{y^+,pl^-} + CI_{y^+,y^-}$, $CI_{y^-,pl^+} + CI_{y^-,y^+}$, $CI_{y^+,pl^+} + CI_{y^+,y^+}$ and $CI_{y^-,pl^-} + CI_{y^-,y^-}$, respectively.

Heterogeneity Across Firms: Clustering strength varies widely across industries and ownership types, reflecting differences in market microstructure friction and investor behavior.

Portfolio Analysis: Amplified LHE Effects

Table 8
Portfolio analysis: amplified LH jumps clustering effects.

	CI	Mkt Cap (10^{-6})	Trd Amt (10^{-8})	AR ($*$ 252)	TO ratio	Std Dev ($*\sqrt{252}$)	Skew	Kurt	Amihud (10^{6})
min	0.05	9.12	1.69	21.6%	26.91	0.51	0.08	4.88	80.23
2	0.06	9.05	1.61	19.1%	28.77	0.51	0.09	5.00	98.79
3	0.07	11.42	1.59	18.0%	27.86	0.51	0.09	5.16	71.76
4	0.08	8.96	1.52	17.5%	29.47	0.51	0.11	5.19	50.55
5	0.09	9.94	1.50	17.2%	30.75	0.51	0.13	5.43	36.24
6	0.10	10.60	1.48	17.0%	31.95	0.50	0.14	5.52	25.51
7	0.12	9.51	1.36	15.5%	34.21	0.49	0.17	5.61	12.50
8	0.15	15.52	1.62	12.4%	37.19	0.49	0.20	5.79	16.36
9	0.20	21.20	1.62	11.0%	40.73	0.47	0.20	5.95	11.62
max	0.32	26.26	1.73	7.8%	37.75	0.44	0.25	6.27	10.67

Note: This table reports various indicators of the portfolios sorted based on the clustering index. Regarding the amplified LH jumps clustering effects, $j \to i$ can be $pl^+ \to pl^+$ or y^+ for upper limits hitting jumps and $pl^- \to pl^-$ or y^-) for lower limits hitting jumps, and CI represents the average value. Mkt Cap denotes the market capital; Trd Amt represents the trading amount; AR represents the average daily return; TO ratio represents the turnover ratio; Std Dev, Skew and Kurt represent the second, third and fourth moments of firms' daily returns; Amihud represents the illiquidity ratio, a lower Amihud value indicates higher liquidity of the stock.

Amplified effect (A): High-CI portfolios exhibit higher expected returns $\to$ LHE is **priced** in the cross-section.

Portfolio Analysis: Reversal LHE Effects

Table 9
Portfolio analysis: reversal LH jumps clustering effects.

	CI	Mkt Cap (10^{-6})	Trd Amt (10^{-8})	AR ($*$ 252)	TO ratio	Std Dev ($*\sqrt{252}$)	Skew	Kurt	Amihud (10^{6})
min	0.02	5.87	1.13	17.6%	27.73	0.52	0.08	4.62	146.96
2	0.03	7.54	1.45	20.3%	29.62	0.52	0.08	4.95	98.69
3	0.04	7.66	1.45	18.1%	30.56	0.52	0.10	5.29	185.06
4	0.04	9.24	1.65	17.7%	30.34	0.51	0.11	5.30	64.10
5	0.05	9.67	1.51	16.9%	29.93	0.50	0.12	5.42	72.69
6	0.07	11.67	1.54	15.8%	31.65	0.50	0.15	5.56	26.17
7	0.08	14.08	1.88	16.3%	33.72	0.49	0.18	5.70	11.71
8	0.11	15.36	1.67	14.2%	38.25	0.48	0.18	5.74	12.06
9	0.16	19.66	1.77	12.0%	37.72	0.47	0.22	5.85	10.92
max	0.29	30.83	1.67	8.0%	36.06	0.44	0.24	6.36	10.49

Note: This table presents various indicators of the portfolios sorted based on the clustering index. Regarding reversal LH jumps clustering effects, $j \to i$ can be $pl^+ \to pl^-$ or y^- for upper limits hitting jumps and $pl^- \to pl^+$ or y^+) for lower limits hitting jumps, and CI represents the average value. Mkt Cap denotes the market capital; Trd Amt represents the trading amount; AR represents the average daily return; TO ratio represents the turnover ratio; Std Dev, Skew and Kurt are the second, third and fourth moments of firms' daily returns; Amihud represents the illiquidity ratio, a lower Amihud value indicates higher liquidity of the stock.

Reversal effect (R): Large CI predicts opposite-direction future jumps, reflecting asymmetric investor reaction.

Portfolio Analysis Summary

- CI helps explain cross-sectional return differences.
- High excitation $\rightarrow$ higher future volatility and return premia.
- LH-related excitation carries information beyond volatility or size.

Further Analysis with Annual LHE Effects

Why Analyze Annual LHE Effects?

Goal: Understand why firms differ in their limit-hitting exciting (LHE) strength.

Motivation

- LHE captures a firm's sensitivity to extreme price movements.
- Different firms (size, liquidity, risk profiles) may react differently to PLH events.
- Annual estimates smooth out noise and reveal structural patterns.

Research Questions

- What firm characteristics drive stronger (or weaker) LHE?
- Does LHE contain predictive information about future return moments?

Determinants of Limit-Hitting Exciting Effects

Table 10
Regression results: determinants of limit-hitting exciting effects.

	(1)	(2)	(3)	(4)
VARIABLES	$CI_{pl^+,R}$	$CI_{pl^-,R}$	$CI_{pl^+,A}$	$CI_{pl^-,A}$
AC	0.139	-0.0387	0.0939	0.202
MC	-1.119***	-1.154***	-0.119	0.779**
InsH	-2.281*	-6.454***	0.976	-2.489
Trd Amt	1.082***	1.433***	1.638***	0.177
Mkt Cap	0.222	0.178	-1.150**	-0.639
AR	-0.0819***	-0.119**	-0.0793**	-0.166***
Std Dev	30.29	-96.96***	-50.00**	-25.38
Skew	-0.799***	-0.966*	-1.030***	-0.381
Kurt	0.162*	0.0224	0.228**	0.113
λ_0^{pl}	-0.0500***	-0.00504	0.0557***	-0.0294***
λ_0^{y}	0.476***	0.842***	0.497***	0.500***
Constant	-15.07***	-8.541	1.381	34.64***
# of Obs	34,091	34,091	34,091	34,091
R^2	6.7%	6.2%	5.4%	2.9%

Key Insights

- Higher trading activity $\rightarrow$ stronger LHE (attention-driven amplification).
- Greater media coverage $\rightarrow$ weaker LHE (improved transparency).
- High small-jump intensity (λ_0^y) $\rightarrow$ weaker LHE (already volatile).
- High LH baseline intensity (λ_0^{pl}) $\rightarrow$ stronger LHE (PLH-prone firms).

Explaining Future Return Moments

Table 11
Regression results: explaining future return moments.

	(1)	(2)	(3)	(4)
VARIABLES	AR	Std Dev	Skew	Kurt
L.$CI_{pl^+,R}$	-0.266**	-0.0433**	-0.0520***	0.123***
L.$CI_{pl^-,R}$	-0.355***	-0.0141	0.00385	0.00713
L.$CI_{pl^+,A}$	-0.179**	-0.0923***	-0.0308***	0.157***
L.$CI_{pl^-,A}$	-0.0848	-0.0489***	-0.00867	0.0721**
L.AC	-0.258***	-0.105***	-0.0194***	-0.0598***
L.MC	1.687***	0.165***	0.0676***	-0.614***
L.InsH	7.147***	-0.194***	0.0913***	1.515***
L.Trd Amt	-0.334***	-0.118***	-0.0917***	0.217***
L.Mkt Cap	-1.966***	-0.0394***	0.0505***	0.311***
L.AR	-0.289***	0.0450***	-0.00450***	-0.0412***
L.Std Dev	0.426***	0.138***	-0.073***	-0.115***
L.Skew	1.123***	0.250***	-0.0290***	-0.475***
L.Kurt	0.0862***	0.0130***	0.00702***	0.0301***
L.λ_0^{pl}	0.416***	-0.0101	-0.0758***	0.0845
L.λ_0^{y}	3.562***	1.092***	-0.115***	-1.606***
Constant	30.87***	5.183***	1.363***	-3.072***
# of obs	29,985	29,985	29,985	29,985
R^2	22.6%	20.7%	9.4%	18.2%

Interpretation

- High CI predicts lower future returns (negative AR coefficients).
- LHE significantly explains future volatility (Std Dev), skewness, and kurtosis.
- LH-based excitation contains **forward-looking jump risk information**.

Why Should LHE Predict Future Return Moments?

Mechanism

- LHE captures latent jump risks not reflected in volatility.
- High LHE firms have stronger **tail risks** and imbalance-driven moves.
- Return moments (skewness, kurtosis) are directly affected by jump activity.

Implications

- LHE is a **forward-looking** measure of jump risk.
- Predictive power beyond standard characteristics (size, turnover, volatility).

Further Analysis with Monthly LHE Estimates

Why Monthly LHE Effects?

Annual LHE: captures long-run structural patterns. **Monthly LHE**: captures short-term dynamics and investor sentiment.

Reasons for Monthly Modeling

- PLH events may cluster in short windows $\rightarrow$ monthly resolution detects cycles.
- LHE strength may vary with market regimes and sentiment.
- Enables monthly portfolio rebalancing $\rightarrow$ out-of-sample tests of LHE pricing.

Constructing Monthly Clustering Index

Monthly CI is computed using the same formula:

$$CI_{j,i} = \frac{\alpha_{j,i}}{\beta_{j,i}}$$

Differences from Annual Estimates

- Estimated using rolling monthly windows.
- More sensitive to sentiment shocks.
- Allows detection of short-lived structural changes.

Out-of-Sample Portfolio Performance (Monthly CI)

Table 12
Portfolio analysis: out-of-sample results.

	CI	EW Return (%)	VW Return (%)	Mkt Cap	Price
min	0.108	1.980	1.372	7.8	16.8
2	0.139	1.928	1.215	7.5	15.7
3	0.160	1.883	1.146	7.7	15.1
4	0.179	1.731	1.153	8.8	15.7
5	0.198	1.613	0.881	9.6	15.3
6	0.220	1.568	0.870	10.2	14.8
7	0.249	1.590	1.312	11.9	14.8
8	0.288	1.616	1.099	15.0	15.1
9	0.350	1.470	0.978	22.6	15.4
max	0.530	1.123	0.478	37.2	17.7
10-1		-0.857 (-2.67)	-0.894 (-1.73)		

Interpretation of CI-Sorted Returns

- CI behaves like a risk factor: higher CI $\rightarrow$ lower future returns.
- 10–1 spread is negative (EW and VW), confirming LHE is **priced**.
- High-CI stocks face greater jump risk $\rightarrow$ require higher compensation.

LHE Effects and Investor Sentiment

Table 13
Regression results: LHE effects and sentiment.

	LHE_A	LHE_R	ULH	LLH
Constant	0.02	-0.13	-0.27	-0.18
Sent	-0.04	0.30***	0.61***	0.40***
AC	0.04	-0.08	-0.11*	-0.15*
MC	0.07	-0.34***	0.08	0.20**
ATV	0.15*	0.19**	0.46***	0.09
ER	-0.15*	0	0.04	0.01
EPU	0.47***	-0.29***	-0.14*	-0.12
# of obs	144	144	144	144
adj. R^2	20.4%	18.7%	44.4%	9.3%

Economic Insights

- Sentiment boosts reversal effects (LHE_R) $\rightarrow$ markets overreact then correct.
- EPU amplifies LHE asymmetry $\rightarrow$ uncertainty increases clustering risk.
- ULH rises when sentiment is strong $\rightarrow$ evidence of herding and

Summary: What Monthly LHE Reveals

Key Takeaways

- Monthly CI is a short-term jump-risk indicator.
- Predicts cross-sectional returns in real time.
- Sensitive to sentiment $\rightarrow$ strong behavioral component.
- Complements annual LHE by capturing regime shifts.

Conclusion

Conclusion

- LH jumps are fundamentally different from small jumps.
- LH events generate strong mutually-exciting effects.
- Proposed jump model significantly outperforms alternatives.
- Market frictions introduced by price limits distort return dynamics.
- Future work: intraday analysis, regime-switching Hawkes, option pricing implications.

Replication

Data Description: Taiwan Market Replication Study

To replicate the empirical analysis of the original paper using the Taiwan stock market, we construct a dataset covering the period from January 1, 2015 to January 1, 2025

Sample selection:

- Include all listed **Taiwan Stock Exchange (TWSE)** common stocks.
- Exclude:
  - TDR and KY-listed foreign companies,
  - Firms listed after January 1, 2023,
  - Stocks placed under full-delivery requirements,
  - Stocks that never experience a limit-hitting (LH) event during the sample period.

Replication: Data Statistics (Taiwan Market)

Metric	Original (China A-Share)	Replication (Taiwan)
# of Stocks	3,646	**820**
Avg. PLH per Year	7.91 (3.14%)	**2.66 (1.06%)**
Max PLH per Year	26.62 (10.56%)	**10.75 (4.26%)**
Min PLH per Year	0.55	**0.1**

Table 1: Comparison of Price-Limit Hitting (PLH) Statistics [2015-2025]

Key Insight

- **Lower Frequency:** PLH events are significantly less frequent in Taiwan (**1.06%**) compared to China (**3.14%**).

Replication: Stock Selection Strategy

Selection Methodology:

- We sorted all 820 sample stocks by their total number of Price-Limit Hitting (PLH) events from 2015 to 2025.

Representative Stocks:

- **High (90th%):**
  6120 (Darwin Precisions)
  $\rightarrow$ **57 events** (High Volatility).
- **Median (50th%):**
  2464 (Mirle Automation)
  $\rightarrow$ **19 events**.
- **Low (10th%):**
  1109 (Hsin Ta Cement)
  $\rightarrow$ **3 events**.

```
總股票檔數: 820
------------------------------
[第 10 百分位]
排名: 82/820
代碼: 6120 達運
次數: 57
------------------------------
[第 50 百分位]
排名: 410/820
代碼: 2464 盟立
次數: 19
------------------------------
[第 90 百分位]
排名: 738/820
代碼: 1109 信大
次數: 3
```

Fig: Ranking stocks by PLH frequency.

Model Specification (1/4): Volatility Dynamics

1. Exogenous Volatility

We model the volatility v_t as:

$$dv_t = \kappa(\theta - v_t)dt + \sigma_v \sqrt{v_t} dW_t^v \quad \text{(Variance)}$$

- v_t: **Realized Volatility** (Pre-calculated from historical data).
- We treat v_t as exogenous to focus on jump excitation.

Model Specification (2/4): Jump Intensity

2. Mutually-Exciting Jump Intensity

The probability of a jump occurring at time t depends on past jumps:

$$\lambda_t^i = \lambda_0^i + \sum_{t_k < t} \alpha_{j,i} \cdot e^{-\beta(t-t_k)} \tag{1}$$

where $i, j \in \{LH \text{ (Limit-Hit)}, S \text{ (Small Jump)}\}$.

Parameter Interpretation:

- λ_0^i (Baseline Intensity): Represents the fundamental probability of a jump.
- $\alpha_{j,i}$: The excitation level (how much a past jump triggers a new one).

Model Specification (3/4): Small Jump Size

1. Small Jumps (S): Normal Distribution

Representing routine market noise or minor shocks.

$$Z^{(S)} \sim \mathcal{N}(\mu_s, {\sigma_s}^2) \tag{2}$$

- μ_s: Average size of small jumps (usually close to 0).
- σ_s: Variability of small jumps.

Model Specification (4/4): Limit-Hit Size

2. Limit-Hitting Jumps (LH): Double Exponential

Following Kou (2002), LH events capture extreme tail risks:

$$f_{LH}(x) = p_{up} \cdot \eta_1 e^{-\eta_1 x} \mathbb{I}_{\{x>0\}} + (1 - p_{up}) \cdot \eta_2 e^{\eta_2 x} \mathbb{I}_{\{x<0\}} \tag{3}$$

- p_{up}: Probability of the limit hit being **Upward**.
- η_1, η_2: Decay rate of the jump tails.
  (Lower η implies "fatter tails" or larger extreme moves).

** These parameters correspond directly to the MCMC outputs shown in the next slide.*

Heterogeneity Analysis (1/2): Estimation Outputs

High PLH: 6120

(57 Limit-Hitting Events)

```
達運參數估計結果 (後 3000 次平均):
mu          -0.081576
kappa        3.170653
theta        0.000863
sigma_v      0.065028
rho         -0.001783
p_up         0.509134
eta1        21.816665
eta2        26.874228
mu_s         0.002815
sigma_s      0.082166
lam0_lh      1.251362
lam0_s      13.354145
a_LL         0.469652
a_LS         0.190143
a_SS         0.320473
beta         0.986654
dtype: float64
```

- Note $\alpha_{LL} \approx 0.47$ (High)
- Note $\eta_1 \approx 21.8$ (Thin Tail)

Low PLH: 1109

(3 Limit-Hitting Events)

```
信大參數估計結果 (後 3000 次平均):
mu          -0.000858
kappa        0.028646
theta        0.010448
sigma_v      0.044845
rho          0.000847
p_up         0.501760
eta1        16.191623
eta2        15.914487
mu_s         0.004866
sigma_s      0.045570
lam0_lh      0.067853
lam0_s      12.853142
a_LL         0.117078
a_LS         0.012096
a_SS         0.117597
beta         1.002881
dtype: float64
```

- Note $\alpha_{LL} \approx 0.11$ (Low)
- Note $\eta_1 \approx 16.2$ (Fat Tail)

Heterogeneity Analysis (2/2): Market Intuition

Param	6120 (Hot)	1109 (Cold)
α_{LL}	**0.47** (Strong)	0.11 (Weak)
η_1	**21.8** (Thin Tail)	**16.2** (Fat Tail)

1. **Sentiment (α)**
   - **6120 (Tech):**
   - High α = **Herding**.
   - Driven by retail momentum.
   - Clustering effect is obvious.

2. **Information (η)**
   - **1109 (Cement):**
   - Low η = **Big Surprise**.
   - Requires massive shock to hit limit.
   - "Real" Information.

Visualizing Excitation: Intensity Dynamics

Panel Analysis:

- **1. Middle (λ_{LH}):** Intensity spikes stack up during the crash. $\rightarrow$ Captures **Panic Selling** (Accumulated Fear).

- **2. Bottom (λ_S):** Small jump intensity fluctuates with a different baseline (Noise).

Extension

Extended Analysis: Taiwan Market (I)

1. **Attention and Disposition Stocks**
   - Taiwan's "attention" and "disposition" stocks are subject to trading restrictions and heightened market scrutiny.
   - These stocks are intuitively more likely to experience limit-hitting (LH) events, and their jump dynamics may reveal stronger self- or cross-excitation patterns.

2. **Index-Level Analysis: TAIX and TAIFEX Index Futures**
   - Extend the model to the Taiwan Stock Index (TAIEX) and Taiwan Index Futures.
   - LH occurrences are relatively rare for indices, but event-based analysis may still capture systematic jump interactions during market stress.

Extended Analysis: Taiwan Market (II)

3. Options Market Extension: Taiwan Index Options

- Analyze how LHEs influence the implied volatility, skewness, and jump-risk premium embedded in Taiwan index options.
- Allows examination of whether jump clustering spills over from the stock market into the derivatives market.

4. Evaluating LHE-Based Factors

- Construct LHE-related factors, such as:
  - estimated jump intensities,
  - excitation components from the Hawkes structure,
  - LH-frequency or LH-duration measures.
- Test their predictive power and performance in asset-pricing or trading frameworks.

Thank you!